Rahul Kumar

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Introduction

Robotics & Automation Engineering graduate with hands-on experience developing autonomous robotic systems using ROS2, SLAM, Nav2, computer vision, and embedded platforms. Experienced in designing, integrating, and validating complete hardware–software solutions on physical robots, with strong programming skills in C++, Python, and Linux.

Education

Symbiosis Institute of Technology, Btech in Robotics and Automation — August 2022 – June 2026

- CGPA: 7.7/10

Projects

Autonomous Differential Drive Robot (Jan 2025 – Apr 2025)

- Designed and built a differential-drive mobile robot from scratch, integrating a Raspberry Pi 4B, RPLIDAR A1M8, planetary geared DC motors with encoder feedback, custom PDS board, and L298N motor drivers. Developed the complete ROS2 stack by creating the URDF, configuring ros2_control, integrating SLAM Toolbox and Nav2, and tuning localization and navigation for reliable autonomous operation on physical hardware.

Pandoobi – Underwater Robot (Aug 2024 – Jan 2025)

- Designed and built a semi-tethered underwater exploration robot from scratch, integrating Raspberry Pi 5, four 920 KV BLDC thrusters, a custom PDS, waterproof camera enclosure, MPU6050 IMU, and BMP280 sensor. Engineered embedded control, live video streaming, and stable underwater navigation.

Uniball Self-Balancing Robot (Jan 2025 – Feb 2025)

- Developed a self-balancing spherical robot using an ESP32, MPU6050 IMU, N20 geared motors with encoder feedback, and a custom chassis. Implemented Kalman filter-based sensor fusion and real-time balancing control for stable directional movement.

Offline AI Assistant (Sep 2025 – Feb 2026)

- Developed an offline edge AI assistant using OpenWakeWord, TinyWhisper, and locally hosted Llama/Mistral LLMs. Integrated wake-word detection, speech recognition, LLM inference, and text-to-speech into a low-latency conversational pipeline.

Computer Vision-Based Home Automation (Aug 2024 – Oct 2024)

- Implemented a computer vision-based home automation system using OpenCV, YOLOv8, and ESP32. Developed real-time human detection for automated control of lights, fans, and power sockets, while enabling gesture-based fan speed adjustment.

EXPERIENCE

Embedded Systems Intern — Pratham Embedded Solutions (Jan 2026 – Mar 2026)

- Built a manually-operated cargo transport robot for moving goods between points, using a repurposed electric-bike motor with throttle-based speed control and handlebar-style steering on a single drive motor.Integrated a linear Lidar sensor for real-time obstacle detection to prevent collisions during operation

Research Assistant — National Insurance Academy (NIA), Pune(Apr 2026 – Current)

- Supporting a World Bank-funded consulting study, "Multi-Hazard Risk Financing Strategy for Assam" under AIRBMP, contributing quantitative risk-loss analysis and financing-instrument research for the project's deliverable reports.

Hackathons and Competitions

Robocon 2024 | Reached Round 3 (Top 30/120+ teams) — Nov 2023 – Aug 2024

Developed autonomous and remote-controlled robots for a seed plantation challenge using Raspberry Pi 5, Jetson Nano, and laser-based navigation.

Robofest 2025 | Grand Finalist (National Level) — Aug 2024 - Jan 2025

Built Pandoobi, an underwater robot for aquatic navigation.

Skills

Languages: C#, C++, SQL, Bash/shell script, Python, Matlab/Simulink

Technologies: Raspberry Pi, Arduino, ROS2, OpenCV, ESP, Machine Learning, TensorFlow, 360 Lidar, PLC programming

Soft skills : Team Coordination, Public Speaking, Time Management, Technical Adaptability, Self-Learning